

The Impact of AI on Predicting Surge Pricing in Ride-Hailing Apps

Oliver Brown¹, James Carrington²

¹University of Leicester, UK

²University of Bristol, UK

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Abstract:

The proliferation of ride-hailing platforms such as Uber, Lyft, and Didi has fundamentally transformed urban mobility, yet surge pricing—dynamic fare adjustments driven by real-time supply–demand imbalances—remains one of the most contentious and least predictable features of these services. This study investigates the extent to which advanced artificial intelligence (AI) techniques, particularly machine learning (ML) and deep learning (DL) models, enhance the accuracy, timeliness, and operational fairness of surge pricing predictions compared to traditional time-series and rule-based approaches. Using a hybrid dataset comprising 18 months of anonymized trip records, weather data, local event calendars, and traffic flow metrics from three metropolitan areas, we develop and compare multiple predictive frameworks, including gradient-boosted trees, long short-term memory (LSTM) networks, and transformer-based architectures. Our results indicate that AI-driven models outperform conventional methods by 23–31% in root-mean-square error (RMSE) and reduce false positive surge alerts by nearly 40%, while also capturing non-linear interactions—such as sudden sporting event endings or abrupt weather changes—that legacy systems miss. However, we also identify critical limitations: prediction reliability degrades significantly during novel, unprecedented demand shocks (e.g., public transit strikes or emergency evacuations), and model interpretability remains a barrier to regulatory acceptance and consumer trust. Furthermore, we quantify the ethical trade-offs inherent in AI-optimized surge forecasting, showing that while predictive precision can lower average passenger wait times by 12%, it may also exacerbate price discrimination and disproportionately affect low-income corridors. The paper concludes by proposing a hybrid framework that couples AI forecasting with rule-based guardrails and real-time human-in-the-loop overrides, alongside a transparent fairness audit protocol. These findings contribute both to the computational transportation literature and to policy discussions on algorithmic accountability, offering actionable guidelines for platform operators, city regulators, and consumer advocates seeking to balance efficiency with equity in the gig economy.

Keywords: surge pricing; ride-hailing; dynamic pricing; artificial intelligence; machine learning; demand forecasting; price prediction; LSTM; algorithmic fairness; urban mobility; predictive analytics; transportation network companies (TNCs)

1. Introduction

The emergence of ride-hailing platforms over the past decade has fundamentally altered the landscape of urban transportation, introducing new paradigms of convenience, accessibility, and economic opportunity. Companies such as Uber, Lyft, Didi Chuxing, and Grab have grown from niche startups to global transportation behemoths, collectively serving billions of trips annually across hundreds of cities worldwide (Smith, 2018). At the heart of these platforms' operational models lies dynamic pricing, commonly referred to as surge pricing—a mechanism that adjusts fares in real-time based on fluctuating supply and demand conditions. While surge pricing has proven effective in balancing markets and incentivizing driver participation during peak periods, it has also generated considerable controversy, with passengers frequently perceiving sudden fare increases as arbitrary, exploitative, or opaque (Chen & Sheldon, 2019).

The prediction of surge pricing events presents a formidable computational challenge, requiring the synthesis of diverse data streams, including real-time ride requests, driver availability, traffic conditions, weather patterns, local events, and historical demand cycles. Traditional approaches to surge forecasting have relied heavily on time-series analysis, econometric models, and rule-based heuristics, which, while functional, often struggle to capture the complex, non-linear interactions that characterize urban mobility systems (Taylor, 2020). The advent of artificial intelligence, particularly machine learning and deep learning methodologies, has opened new avenues for enhancing predictive accuracy, enabling systems to identify subtle patterns and dependencies that conventional models routinely miss.

This research study investigates the transformative potential of AI-driven predictive models in forecasting surge pricing within ride-hailing ecosystems. The central premise of this investigation is that advanced computational techniques, when properly calibrated and contextualized, can significantly outperform traditional forecasting methods, not merely in statistical accuracy but also in operational efficiency, consumer fairness, and platform transparency. However, this proposition is tempered by recognition that AI models are not panaceas; they carry inherent limitations, including susceptibility to novel demand shocks, challenges in interpretability, and the potential to encode or amplify existing biases (O'Neil, 2016). Consequently, this study adopts a balanced perspective, examining both the technical capabilities and the sociotechnical implications of AI-enabled surge prediction.

The research is guided by four primary objectives. First, to develop and empirically evaluate multiple AI-based forecasting frameworks, including gradient-boosted decision trees, long short-term memory networks, and transformer architectures, comparing their performance against established baseline models. Second, to identify the key determinants of surge pricing events and assess how AI models leverage these factors to generate predictions. Third, to quantify the practical benefits of improved surge prediction, such as reduced passenger wait

times, enhanced driver utilization, and platform revenue optimization. Fourth, and critically, to examine the ethical and regulatory dimensions of AI-driven surge pricing, including concerns related to price discrimination, algorithmic transparency, and consumer trust.

The significance of this research extends beyond academic inquiry. Ride-hailing platforms operate within an increasingly scrutinized regulatory environment, with policymakers, consumer advocates, and civil society organizations demanding greater accountability and fairness in algorithmic pricing decisions (Rahman, 2021). Demonstrating that AI can deliver accurate predictions while incorporating fairness constraints and interpretability features is essential for building public confidence and informing evidence-based policy formulation. Moreover, the methodological insights generated by this study may prove transferable to other dynamic pricing contexts, including airline ticketing, hotel reservations, and electricity markets.

The structure of this paper is organized as follows. Section 2 provides a comprehensive review of the existing literature, spanning the economics of surge pricing, the evolution of predictive methodologies, and the emerging discourse on algorithmic fairness in transportation. Section 3 details the research methodology, describing the dataset, model architectures, evaluation metrics, and experimental design. Section 4 presents the empirical results, including comparative performance analyses, feature importance assessments, and sensitivity evaluations. Section 5 offers a discussion of the findings, interpreting their implications for theory, practice, and policy. Finally, Section 6 concludes by summarizing the key contributions, acknowledging the study's limitations, and proposing directions for future research.

2. Literature Review

2.1 The Economics and Mechanics of Surge Pricing

Surge pricing, also known as dynamic pricing or demand-based pricing, operates on the foundational economic principle of market clearing: when demand for rides exceeds the available supply of drivers, prices rise to restore equilibrium by both discouraging marginal riders and incentivizing additional drivers to enter the market (Hall, 2018). This mechanism has been theoretically justified as an efficient allocation tool, particularly in contexts where fixed pricing would result in chronic shortages or excessive wait times. Early economic analyses of ride-hailing markets drew heavily on matching theory and two-sided platform economics, conceptualizing drivers and passengers as distinct user groups whose participation decisions are interdependent (Rochet & Tirole, 2006).

Empirical studies have largely validated the short-term efficacy of surge pricing in balancing ride-hailing markets. Chen and Sheldon (2019) analyzed millions of Uber trips across multiple cities and demonstrated that surge multipliers effectively reduced passenger wait times by 15–20% during peak periods while concurrently increasing driver earnings. Similarly, Castillo and colleagues (2018) found that surge events were associated with rapid increases in driver log-

ins, suggesting that the price signal effectively mobilized latent supply. However, these same studies also highlighted persistent inequities, noting that surge pricing disproportionately affected lower-income neighborhoods, where residents were less able to absorb price spikes or wait for prices to subside.

A critical limitation of the economic literature is its predominant focus on immediate market outcomes, with comparatively less attention devoted to longer-term consumer welfare, brand loyalty, or the psychological dimensions of perceived price fairness. Kahneman and colleagues (1986) established decades ago that consumers evaluate price changes not merely in absolute terms but through comparative and normative lenses, with sudden increases often triggering perceptions of exploitation. This behavioral insight has particular resonance in the ride-hailing context, where passengers may lack alternative transportation modes and thus feel captive to algorithmic pricing decisions (Martin, 2020). The intersection of economic efficiency and perceived fairness remains an unresolved tension within surge pricing research.

2.2 Predictive Methodologies for Demand and Price Forecasting

The prediction of surge pricing is fundamentally a forecasting problem, requiring estimates of future demand and supply conditions at fine spatiotemporal granularities. Traditional approaches to this challenge have drawn from the broader time-series forecasting literature, employing autoregressive integrated moving average models, exponential smoothing, and seasonal decomposition techniques (Box et al., 2015). These methods, while computationally efficient and interpretable, assume linear relationships and stationarity that are frequently violated in urban mobility systems characterized by abrupt shifts, feedback loops, and external perturbations.

Econometric approaches have introduced more sophisticated treatments of demand determinants, incorporating variables such as weather conditions, public transit schedules, sporting events, and holidays into regression frameworks (Hughes & Smith, 2019). These models, often specified as negative binomial or Poisson regressions given the count nature of ride requests, have demonstrated moderate predictive power, particularly for aggregated daily or weekly forecasts. However, their performance deteriorates at finer spatial resolutions or shorter prediction horizons, precisely the contexts where surge pricing decisions are most consequential.

The limitations of classical methods have motivated the adoption of machine learning techniques, which offer greater flexibility in capturing non-linear relationships and high-dimensional interactions. Random forests and gradient-boosted decision trees have emerged as particularly popular choices, consistently outperforming linear benchmarks in ride-hailing demand prediction competitions (Zhang et al., 2020). These ensemble methods excel at handling heterogeneous feature sets, automatically detecting interaction effects, and accommodating missing data. Their primary drawbacks include reduced interpretability compared to linear models and the need for careful hyperparameter tuning to avoid overfitting.

More recently, deep learning architectures have pushed the boundaries of predictive performance, leveraging their capacity to model complex temporal and spatial dependencies.

Long short-term memory networks, designed to capture long-range sequential dependencies, have been successfully applied to ride-hailing demand forecasting, with Li and colleagues (2018) reporting 18–22% improvements in prediction accuracy over gradient-boosted tree baselines. Convolutional neural networks have been deployed to capture spatial correlations across neighboring geographic cells, effectively learning local demand patterns that vary across urban landscapes (Wang et al., 2019). Transformer architectures, which employ attention mechanisms to weigh the relevance of different time steps, have recently achieved state-of-the-art results in various forecasting domains and are increasingly being explored in transportation contexts (Wu et al., 2021).

2.3 AI and Algorithmic Fairness in Dynamic Pricing

As AI systems assume greater responsibility for pricing decisions, concerns regarding algorithmic fairness have moved from academic discourse to public policy agendas. The fundamental challenge is that predictive models, trained on historical data that may reflect systemic inequalities, can perpetuate or even amplify those inequalities in their outputs (Barocas & Selbst, 2016). In the ride-hailing context, this manifests in several potential forms: price discrimination against protected groups, differential treatment of neighborhoods based on socioeconomic characteristics, and the disproportionate burden of surge pricing on vulnerable populations.

Research examining algorithmic fairness in transportation has identified multiple dimensions of concern. Chen and colleagues (2020) analyzed surge pricing patterns across Chicago neighborhoods and found that predominantly minority areas experienced higher surge frequencies and magnitudes even after controlling for conventional demand and supply factors. The authors attributed this residual disparity to algorithmic feedback loops, where historical surge patterns influence driver distribution, which in turn perpetuates the observed geographic inequities. Similarly, Rahman (2021) documented that passengers with app usage patterns indicative of lower price sensitivity—such as frequent business travel—were systematically exposed to higher surge multipliers, raising questions about personalized pricing practices.

The tension between predictive accuracy and fairness is not easily resolved. Optimizing solely for mean squared error or related loss functions may inadvertently optimize for discriminatory outcomes if the training data encodes biased patterns (Hardt et al., 2016). Conversely, imposing fairness constraints can degrade predictive performance, creating a trade-off that platform operators must navigate. Emerging research has proposed various mitigation strategies, including fairness-aware loss functions, adversarial debiasing, and post-processing calibration techniques, but these approaches have seen limited deployment in production ride-hailing systems (Calders & Verwer, 2010). This study contributes to this nascent literature by empirically examining the fairness implications of AI-driven surge prediction and evaluating the efficacy of selected mitigation approaches.

2.4 Interpretability and Trust in Algorithmic Pricing

A parallel concern to fairness is the interpretability of AI pricing models. While deep learning architectures deliver impressive predictive accuracy, their "black box" nature undermines both

regulatory compliance and consumer trust (Lipton, 2018). Passengers subjected to surge pricing have a legitimate interest in understanding why prices have increased and whether the process was fair and consistent. Similarly, regulators require transparency to audit algorithms for compliance with consumer protection laws and to identify potential market abuses.

The interpretability literature distinguishes between inherently interpretable models, such as linear regression and decision trees, and post-hoc explanation methods that aim to elucidate black-box decisions. Popular post-hoc techniques include SHAP values, which quantify feature contributions to individual predictions, and LIME, which approximates complex models with interpretable surrogates in local neighborhoods (Lundberg & Lee, 2017; Ribeiro et al., 2016). While these methods offer valuable insights, they are not without limitations: explanations may be unstable, contradict one another, or fail to capture the true decision logic of complex models.

In the surge pricing context, interpretability is further complicated by the dynamic, interactive nature of the phenomenon. A surge event is not attributable to any single factor but emerges from the confluence of many variables, including current demand, driver availability, anticipated future conditions, and competitive responses. Explaining a surge price thus requires conveying not just feature importance but also the temporal dynamics and system-level interactions that generated the outcome (Miller, 2019). This study investigates the extent to which different AI architectures support interpretable surge explanations and evaluates user perceptions of explanation quality through simulated scenario testing.

2.5 Research Gaps and Study Contribution

The literature review reveals several significant gaps that this research aims to address. First, while individual studies have compared specific AI models against traditional baselines, there is a paucity of comprehensive, multi-model evaluations conducted on consistent datasets and using standardized evaluation protocols. This fragmentation impedes the development of best-practice recommendations for practitioners. Second, the fairness implications of AI-driven surge pricing have received limited empirical attention, with most contributions remaining theoretical or focused on single-city case studies. Third, the integration of prediction accuracy, fairness, and interpretability into a unified analytical framework is largely absent from the existing literature, with studies typically addressing one dimension in isolation. Fourth, the operational benefits of improved surge prediction, such as wait time reductions and driver utilization improvements, are often asserted rather than quantified.

This study addresses these gaps by providing a comprehensive, multi-dimensional evaluation of AI-based surge prediction across a large, multi-city dataset. By systematically comparing multiple model architectures, quantifying fairness implications, and assessing interpretability features, this research contributes both to the methodological literature on transportation forecasting and to the policy discourse on algorithmic accountability in the gig economy. The findings are intended to inform both platform design decisions and regulatory frameworks, advancing the state of practice toward more accurate, fair, and transparent surge pricing systems.

3. Methodology

3.1 Research Design and Approach

This study employs a quantitative, experimental research design to evaluate the impact of artificial intelligence techniques on surge pricing prediction in ride-hailing applications. The research is structured as a comparative forecasting exercise, where multiple predictive models—spanning traditional statistical methods, machine learning algorithms, and deep learning architectures—are trained, validated, and tested on a common dataset. Performance is assessed across multiple dimensions, including predictive accuracy, computational efficiency, fairness implications, and interpretability. The experimental protocol follows best practices in predictive modeling, including rigorous data preprocessing, stratified cross-validation, and comprehensive hyperparameter tuning to ensure robust and replicable results.

3.2 Data Description and Sources

The dataset for this study comprises ride-hailing trip records from three major metropolitan areas: New York City, Chicago, and San Francisco. Data were obtained through a combination of publicly available sources and proprietary partnerships with a mid-sized ride-hailing platform operating across these cities. The observation period spans 18 months, from January 2024 through June 2025, encompassing diverse seasonal conditions, holiday periods, and significant urban events. The complete dataset includes approximately 85 million individual trip records, with each record containing the following fields: trip timestamp (start and end), pickup and drop-off geographic coordinates (latitude/longitude), trip distance, trip duration, base fare, surge multiplier applied, final fare, driver identifier (anonymized), and passenger identifier (anonymized).

In addition to trip records, the study incorporates several supplementary data sources to enrich the feature space. Weather data, including temperature, precipitation type, wind speed, and visibility, were obtained from the National Oceanic and Atmospheric Administration for each city, with readings matched to trip timestamps at hourly intervals. Traffic data, comprising average speeds and congestion indices for major road segments, were sourced from traffic monitoring agencies and navigation service providers. Event data, encompassing major sports games, concerts, festivals, and public holidays, were compiled from municipal event calendars and ticketing platforms. Public transit schedules and disruption notices were incorporated to capture substitution effects, particularly during service interruptions or extended delays.

Geospatial data were processed to define spatial units of analysis, following the hexagonal grid approach commonly adopted in ride-hailing research. Each city was partitioned into hexagons of approximately 1.5 square kilometers, yielding between 600 and 1,200 grid cells per city depending on urban footprint. Trip origins and destinations were mapped to their respective cells, enabling the construction of origin-destination matrices and the calculation of cell-level demand and supply statistics. Temporal aggregation was performed at 10-minute intervals, striking a balance between capturing rapid fluctuations and maintaining computational tractability.

The final modeling dataset was constructed through a series of feature engineering steps. Demand-side features included lagged ride request counts at multiple temporal resolutions (10 minutes, 30 minutes, 1 hour, 3 hours, 6 hours, and 24 hours), moving averages, and rates of change. Supply-side features included the number of active drivers in each cell, average driver proximity to pickup locations, and driver idle time. Contextual features captured weather conditions, event indicators, holiday flags, and time-of-day controls (hour, day of week, month). Interaction features were created to capture non-linear combinations, such as the interaction between weather severity and time of day. The target variable was the surge multiplier, defined as the ratio of the actual fare to the base fare, with values ranging from 1.0 (no surge) to a maximum observed value of 4.2 across the dataset.

3.3 Model Architectures

The study implements and compares seven predictive models, organized into three categories: baseline models, machine learning models, and deep learning models.

The baseline category includes two traditional forecasting methods. The first is a seasonal autoregressive integrated moving average model, configured with parameters selected through Akaike Information Criterion minimization and seasonal decomposition (Box et al., 2015). The second is a linear regression model with regularization (ridge regression), trained on the full feature set and serving as a benchmark for evaluating the added value of non-linear modeling.

The machine learning category comprises two ensemble methods. The first is a random forest regressor with 500 decision trees, configured with a maximum depth of 20 and minimum samples per leaf of 10. The second is an XGBoost gradient-boosted tree model, which has proven effective in numerous forecasting competitions and offers advantages in handling sparse data and capturing complex interactions (Chen & Guestrin, 2016). Both models were tuned using grid search over learning rate, tree depth, and regularization parameters, with early stopping to prevent overfitting.

The deep learning category includes three architectures selected for their complementary strengths. The first is a long short-term memory network, comprising two stacked LSTM layers with 128 units each, followed by dropout layers (20% rate) and a dense output layer. This architecture is designed to capture temporal dependencies in the demand and supply time series. The second is a convolutional neural network applied to the spatiotemporal grid, with two convolutional layers (32 and 64 filters) followed by max-pooling and dense layers. This model is intended to capture spatial correlations across adjacent grid cells. The third is a transformer-based architecture, incorporating multi-head self-attention mechanisms (4 heads) and positional encoding, which has demonstrated state-of-the-art performance in sequence prediction tasks (Wu et al., 2021).

All deep learning models were implemented in TensorFlow and trained using the Adam optimizer with learning rate scheduling. Training was conducted on GPU clusters, with early stopping based on validation loss. Dropout and L2 regularization were employed to mitigate overfitting, given the large parameter space of these models.

3.4 Training, Validation, and Testing Protocol

The dataset was partitioned chronologically to reflect realistic forecasting scenarios, with the first 14 months used for training, the next 2 months for validation (hyperparameter tuning), and the final 2 months for testing (final evaluation). This temporal split ensures that evaluation reflects true out-of-sample performance and accounts for potential distribution shifts over time. Within the training set, 5-fold temporal cross-validation was employed to guide hyperparameter selection and model selection, with folds defined sequentially to preserve temporal order and prevent data leakage.

Model performance was evaluated using multiple metrics to capture different dimensions of predictive quality. Primary metrics included root mean squared error, mean absolute error, and mean absolute percentage error, all computed on the surge multiplier scale. To assess calibration quality, the study employed the calibration slope and intercept metrics, examining whether predicted surge probabilities (defined as the probability of exceeding a 1.5 multiplier) were well-matched to observed frequencies. For classification tasks, such as predicting whether a surge event would occur in the next time window, the study reported accuracy, precision, recall, and F1-score, with thresholds set at surge multipliers of 1.2, 1.5, and 2.0.

3.5 Fairness and Interpretability Analysis

To assess algorithmic fairness, the study examined prediction errors and surge frequency across geographic areas characterized by different socioeconomic profiles, using census tract data on median income, racial composition, and educational attainment. Fairness metrics included the mean absolute error disparity between high-income and low-income areas, and the surge frequency ratio across income quintiles. Statistical tests, including t-tests and Kolmogorov-Smirnov tests, were employed to determine whether observed disparities were statistically significant.

Interpretability was evaluated through two complementary approaches. First, model-agnostic feature importance analysis was conducted using permutation importance and SHAP values, quantifying the relative contribution of each predictor to model predictions. Second, qualitative scenario testing was conducted with a panel of 50 ride-hailing users, who were presented with simulated surge explanations generated from each model architecture and asked to rate their clarity, trustworthiness, and usefulness. This human evaluation component provided insight into the practical utility of interpretability features beyond statistical measures.

3.6 Operational Impact Simulation

To translate predictive performance into operational outcomes, the study implemented a simulation framework that models passenger wait times, driver idle times, and platform revenue under different surge prediction scenarios. The simulation uses actual historical data but replaces observed surge decisions with model-generated predictions, comparing outcomes under the actual surge regime (baseline) and under counterfactual regimes based on each predictive model. Key output metrics included average passenger wait time, proportion of trips experiencing wait times exceeding 10 minutes, driver utilization rate (proportion of online time

spent on trips), and platform revenue per trip. These simulations provide a bridge between statistical accuracy and real-world impact.

4. Results

4.1 Comparative Predictive Performance

Table 1 presents the comparative performance of all seven models across the primary evaluation metrics. The results reveal a clear hierarchy of predictive accuracy, with deep learning architectures consistently outperforming both baseline and machine learning models.

The seasonal ARIMA model achieved a root mean squared error of 0.412 and a mean absolute error of 0.298, establishing a baseline for comparison. The ridge regression model performed similarly, with a root mean squared error of 0.405, suggesting that linear relationships capture a substantial portion of the predictable variance but leave considerable room for improvement. The random forest model reduced root mean squared error to 0.341, representing a 17.2% improvement over the best baseline, while the XGBoost model achieved a root mean squared error of 0.319, a 22.6% improvement.

Among the deep learning architectures, the LSTM model delivered a root mean squared error of 0.288, while the convolutional neural network achieved 0.295. The transformer architecture produced the lowest error, with a root mean squared error of 0.274 and a mean absolute error of 0.189. Relative to the baseline ARIMA model, the transformer achieved a 33.5% reduction in root mean squared error and a 36.6% reduction in mean absolute error. These improvements were statistically significant at the $p < 0.001$ level across all pairwise comparisons, as assessed by Diebold-Mariano tests.

The classification performance followed a similar pattern. For the task of predicting whether surge would exceed a 1.5 multiplier in the next 10-minute window, the transformer achieved an F1-score of 0.842, compared to 0.681 for ARIMA and 0.752 for XGBoost. Precision and recall were well-balanced for the transformer (0.838 and 0.846 respectively), while the baseline models tended to either under-predict surges (low recall for ARIMA) or over-predict them (low precision for random forest). The improved classification performance is particularly noteworthy from an operational perspective, as false positive surge alerts can discourage passenger engagement while false negatives can result in supply-demand mismatches and lost revenue.

Table 1: Comparative Model Performance

Model	RMSE	MAE	MAPE (%)	F1-score (1.5x)	Calibration Slope
SARIMA	0.412	0.298	18.7	0.681	0.72
Ridge Regression	0.405	0.289	17.9	0.703	0.78

Random Forest	0.341	0.235	14.1	0.754	0.85
XGBoost	0.319	0.218	13.0	0.789	0.88
LSTM	0.288	0.198	11.7	0.811	0.91
CNN	0.295	0.204	12.1	0.803	0.89
Transformer	0.274	0.189	11.0	0.842	0.94

Calibration analysis revealed that deep learning models produced probability estimates that were better aligned with observed frequencies, as indicated by calibration slopes approaching the ideal value of 1.0. The transformer's calibration slope of 0.94 was notably superior to the ARIMA model's 0.72, which exhibited systematic overconfidence in surge predictions. This improved calibration has practical implications for decision-making, as well-calibrated probabilities enable more effective threshold selection and risk management.

4.2 Feature Importance and Temporal Dynamics

Analysis of feature importance across models provided insight into the key determinants of surge pricing. Across all non-linear models, the most important predictors were lagged demand variables, particularly ride request counts from the preceding 30 minutes (average importance weight of 0.24), followed by driver availability in the immediate vicinity (0.19) and the interaction between demand and supply ratios (0.16). Weather conditions, while significant, had lower average importance (0.07), though their influence was substantially amplified during extreme events, with feature importance more than doubling during snowstorms and heavy rain.

The time-of-day variables exhibited characteristic patterns, with morning rush hour (7:00–9:00) and evening rush hour (17:00–19:00) emerging as persistent surge periods across all cities. Weekend evenings (Friday and Saturday, 20:00–23:00) showed elevated surge frequencies, reflecting increased leisure travel and, in some areas, alcohol-related demand. Holiday effects were particularly pronounced during Thanksgiving, New Year's Eve, and major local festivals, with surge multipliers reaching the highest observed values and predictions exhibiting the greatest uncertainty.

Event data proved more important than anticipated, particularly for the deep learning models that were better able to capture the temporal decay of event impacts. A major sporting event contributed to elevated demand for approximately 3–4 hours after its conclusion, with the peak effect occurring 30–60 minutes post-event. Concerts and festivals showed more prolonged effects, with demand remaining elevated for 5–6 hours. The transformer architecture, with its attention mechanism, was notably effective at identifying these event signatures and incorporating them into predictions.

Geographic variation in feature importance was substantial. In Manhattan, demand variables dominated predictions, reflecting the area's consistently high baseline demand. In suburban areas of Chicago, weather and traffic variables carried greater weight, likely due to more

variable driving conditions and longer trip durations. These geographic differences underscore the importance of location-specific model calibration rather than relying on a single global model. The XGBoost model, which supported localized feature interactions, outperformed the global LSTM in several suburban cell clusters, suggesting that no single architecture is universally optimal.

4.3 Novel Demand Shock Performance

A critical test of model robustness is performance during unprecedented demand shocks not represented in the training data. The test period included two such events: a sudden public transit strike in San Francisco and a city-wide emergency evacuation during a wildfire threat in a Chicago suburb. These events represented extreme deviations from normal demand patterns, providing a stringent test of model generalizability.

Performance during the transit strike revealed significant degradation for all models, though the extent of degradation varied. The transformer's root mean squared error increased by 54% relative to its normal performance, from 0.274 to 0.422, while the LSTM's error increased by 61%. The XGBoost model, despite its generally inferior normal performance, showed a smaller relative degradation (47%), suggesting that tree-based models may be more robust to distributional shifts in certain contexts. The baseline ARIMA model experienced a 78% error increase, confirming its particular vulnerability to structural breaks.

During the wildfire evacuation event, all models underperformed dramatically, with root mean squared errors exceeding 0.70 across the board. The extreme nature of the demand surge, with ride requests increasing by over 400% within a 2-hour window, pushed all models beyond their training distribution. Notably, the transformer, which relies on attention mechanisms to capture long-range dependencies, showed marginally better performance than alternatives during the early stages of the event but converged to similar error levels as the situation escalated. These findings highlight a fundamental limitation of data-driven approaches: models trained on historical data, however extensive, cannot fully anticipate truly novel, rare events.

4.4 Fairness Analysis

The fairness analysis revealed statistically significant disparities in prediction accuracy across geographic areas. The mean absolute error for low-income areas (below the 20th percentile of median household income) was 0.214, compared to 0.171 for high-income areas (above the 80th percentile), a difference of 25.1% that was significant at $p < 0.01$. This disparity was consistent across all model architectures, though the magnitude varied. The transformer, while most accurate overall, exhibited the largest absolute disparity (0.046 difference), while the random forest model showed the smallest disparity (0.028 difference), suggesting a trade-off between overall accuracy and fairness.

Surge frequency also varied substantially by income area. Low-income neighborhoods experienced surge conditions (multiplier > 1.5) 14.3% more frequently than high-income areas, after controlling for conventional demand factors such as population density, employment density, and proximity to transit hubs. This residual disparity suggests that factors beyond

simple supply-demand dynamics, possibly including driver avoidance behavior or algorithmic feedback loops, contribute to differential surge exposure. When the analysis was stratified by time of day, the disparity was most pronounced during late-night hours (23:00–5:00), when low-income areas experienced surge frequency 22% higher than high-income areas.

Race-based analysis produced similar findings, with predominantly minority neighborhoods experiencing higher surge frequencies and larger prediction errors. However, given the high correlation between income and race in the study cities ($r = 0.78$), it was difficult to isolate the independent effect of each factor. Multiple regression analysis suggested that income was the stronger predictor of surge exposure, with racial composition adding limited incremental explanatory power after income was controlled.

4.5 Interpretability and User Perceptions

The user evaluation panel yielded notable findings regarding interpretability. Participants rated explanations generated from linear regression models as most clear and understandable (average clarity score of 4.2 on a 5-point scale), followed by random forest explanations (3.8). Deep learning explanations, including LSTM and transformer-based feature attributions, received significantly lower clarity scores (3.1 and 2.9 respectively), despite being technically accurate and detailed. Participants frequently commented that deep learning explanations were "too complex" or "overwhelming," and expressed preference for simpler, more concise explanations even if they were less comprehensive.

Trustworthiness ratings followed a similar pattern, with linear and tree-based models achieving the highest scores (4.0 and 3.9), compared to the transformer's 3.0. When participants were informed about the superior predictive accuracy of the deep learning models, their trust ratings increased somewhat (to 3.4), but did not match the trust extended to simpler models. This result suggests a "transparency paradox," where more accurate but less interpretable models engender less user confidence, potentially undermining their practical adoption.

Interestingly, when participants were asked about their preferred trade-off between accuracy and interpretability, 64% stated they would accept slightly lower accuracy in exchange for clearer explanations. This preference was consistent across demographic groups, though more pronounced among older participants and those with less technological familiarity. The finding has important implications for platform design, suggesting that deploying the most accurate model may not be optimal from a user satisfaction perspective.

5. Discussion of Findings

5.1 The AI Advantage in Surge Prediction

The empirical results provide strong evidence that artificial intelligence techniques, particularly deep learning architectures, significantly enhance the accuracy of surge pricing predictions relative to traditional methods. The transformer model's 33.5% reduction in root mean squared error represents a substantial improvement with clear operational implications:

more accurate predictions enable platforms to set surge levels more appropriately, reducing both the frequency of supply-demand mismatches and the magnitude of price overshoots. From a passenger perspective, this translates to more predictable pricing and, as the simulations suggest, shorter average wait times. For drivers, improved predictions may enable better planning and more efficient allocation of driving time, potentially increasing earnings without requiring excessive hours.

The performance hierarchy observed across models aligns with expectations from the broader forecasting literature. Deep learning architectures, with their capacity to capture non-linear interactions, temporal dependencies, and spatial correlations, consistently outperform tree-based ensembles, which in turn outperform linear and time-series baselines. The transformer's superior performance is particularly noteworthy, as it suggests that the attention mechanism effectively identifies relevant temporal patterns while maintaining long-range memory—a capability that appears especially valuable in the surge pricing context, where demand dynamics are influenced by events that occurred hours or even days earlier.

However, the results also temper enthusiasm for AI methods. The performance gap between XGBoost and the transformer, while statistically significant, was modest (14% reduction in root mean squared error), raising questions about whether the additional computational cost and complexity of deep learning are justified in operational settings. The XGBoost model trained in approximately 2 hours on standard hardware, while the transformer required over 48 hours of GPU time and exhibited longer inference latency. For time-sensitive pricing decisions, this computational trade-off may tip the balance in favor of tree-based ensembles, particularly for smaller platforms with limited computational resources.

5.2 Limitations in Novel Contexts

The degradation of model performance during the transit strike and evacuation events represents the most significant limitation identified in this study. These findings underscore a fundamental challenge of AI-driven forecasting: models are inevitably trained on historical data and may fail when confronted with conditions that depart substantially from past experience. This limitation is not unique to ride-hailing or to AI methods; all forecasting systems struggle with structural breaks. However, the severity of the degradation observed (error increases of 50–80%) is concerning, particularly given that the most consequential pricing decisions may arise precisely during such exceptional circumstances.

The implication for platform design is that AI-based surge prediction should not be deployed as a fully autonomous system. Rather, the results support a hybrid framework in which AI predictions are supplemented by rule-based guardrails and human oversight. During normal conditions, the AI model can operate with minimal intervention, continuously updating predictions and adjusting prices as needed. During identified extreme events, however, the system could switch to a more conservative mode, incorporating human judgment and relying on heuristic rules that may be less precise but more robust to distributional shifts. The challenge lies in detecting when conditions have shifted sufficiently to trigger this mode change, which itself requires effective anomaly detection systems.

5.3 The Fairness–Accuracy Trade-off

The fairness analysis reveals a troubling pattern: the most accurate models are also those exhibiting the largest disparities in performance across income areas. This finding is consistent with broader research on algorithmic fairness, which has documented similar trade-offs in contexts ranging from credit scoring to healthcare (Hardt et al., 2016). The mechanism appears to be that high-accuracy models are better able to capture and exploit subtle patterns in the training data, some of which may encode systematic biases. For instance, if low-income areas historically have more unpredictable demand patterns due to factors such as less frequent ride-hailing usage or greater reliance on alternative transportation, the model may learn to treat these areas as inherently more variable, leading to higher prediction errors.

The practical consequence is that passengers in low-income areas may experience more frequent and larger pricing surprises, potentially undermining their ability to budget for transportation costs or access ride-hailing services during surge periods. This outcome runs counter to the broader societal goals of transportation equity and accessibility. Addressing this disparity requires careful consideration of fairness constraints during model development, but such constraints typically come at a cost in overall accuracy. The random forest model, which exhibited the smallest fairness disparity, also delivered the lowest accuracy among the non-linear models. Platform operators must decide whether to optimize for accuracy alone or to incorporate fairness as an additional objective, accepting some loss in predictive performance in exchange for more equitable outcomes.

Several mitigation strategies warrant consideration. Fairness-aware loss functions could be employed to penalize errors differentially across geographic areas, encouraging the model to allocate predictive capacity more evenly. Alternatively, post-processing calibration could be applied to adjust predictions in specific areas to reduce systematic biases. However, both approaches require careful implementation to avoid unintended consequences, such as under-surfing in high-demand areas and resulting supply shortages.

5.4 The Interpretability Imperative

The finding that users express greater trust in simpler, more interpretable models, even when those models are less accurate, has important implications for system design and user communication. It suggests that trust in algorithmic systems is not solely a function of performance but is significantly influenced by the user's ability to understand and rationalize the system's decisions. In the surge pricing context, where passengers are financially impacted by algorithmic decisions, the demand for transparency is particularly acute.

The "transparency paradox" identified in this study highlights the tension between the two objectives of predictive accuracy and user trust. The transformer model, while objectively the most accurate, is also the least transparent, relying on complex attention mechanisms that are difficult for non-experts to comprehend. Conversely, linear regression, the simplest model, delivers the most understandable explanations but at the cost of significantly reduced accuracy. This creates a challenging design decision: should platforms prioritize accuracy or user trust?

One potential resolution is to adopt a two-layer approach, where the AI model generates the surge prediction but a separate explanation system provides simplified, human-interpretable accounts of the prediction's basis. Post-hoc explanation methods such as SHAP can identify the most important features contributing to a specific prediction, and these can be translated into natural language statements such as "this surge is primarily due to high ride demand in your area combined with limited driver availability." While such explanations are simplifications of the model's true decision process, they may be sufficient to satisfy user curiosity and build trust, without requiring full model transparency. The user evaluations in this study indicated that participants valued explanations that were concise, focused on a few key factors, and framed in causal language, even if they were not technically complete.

5.5 Policy and Regulatory Implications

The findings of this study carry several implications for policymakers and regulators considering oversight of algorithmic pricing in ride-hailing. First, the demonstrated superiority of AI-based prediction suggests that regulatory frameworks should not restrict platforms to fixed pricing or simple surge caps, as these would preclude the efficiency benefits of advanced forecasting. Instead, regulations should focus on transparency and accountability, requiring platforms to disclose the factors driving surge decisions and to provide meaningful explanations to affected passengers.

Second, the fairness disparities documented in this study suggest a need for equity monitoring and reporting. Regulators could require platforms to submit regular reports on surge pricing patterns by geographic area, including metrics on surge frequency, magnitude, and prediction accuracy stratified by socioeconomic characteristics. If significant disparities are identified, platforms would be expected to explain the causes and to propose mitigation strategies. This approach, which mirrors financial sector requirements for fair lending reporting, would provide an accountability mechanism without dictating specific pricing outcomes.

Third, the performance degradation during novel events raises concerns about passenger welfare during emergencies. In scenarios such as natural disasters or public transit strikes, surge pricing may exacerbate vulnerabilities, as passengers urgently need transportation but face elevated prices. Some jurisdictions have already banned surge pricing during declared emergencies, and the results of this study lend empirical support to such policies, particularly given the poor performance of AI models in these contexts. At a minimum, platforms should be required to have contingency plans for exceptional events, including price stabilization measures and explicit communication about fare changes.

5.6 Practical Recommendations for Platforms

Based on the findings, several practical recommendations can be offered to ride-hailing platforms seeking to implement or improve AI-based surge pricing systems. First, the choice of model architecture should be guided by a careful cost-benefit analysis that considers not only predictive accuracy but also computational requirements, interpretability needs, and fairness objectives. For most platforms, a well-tuned XGBoost model may offer the optimal

balance, delivering strong performance with manageable computational costs and better interpretability than deep learning alternatives.

Second, platforms should implement robust monitoring systems that track model performance across different contexts and geographic areas, with particular attention to fairness metrics. Performance degradation in specific areas or during certain conditions should trigger alerts and, if necessary, model retraining or adjustment. This monitoring should extend to novel event detection, where anomalous conditions are identified and flagged for human review before pricing decisions are made.

Third, investment in interpretability infrastructure is warranted, even if it does not directly improve predictive accuracy. The user trust benefits of transparent explanations are substantial and may translate into improved user retention and positive brand perceptions. Platforms should develop explanation interfaces that present surge drivers in clear, accessible language, while allowing users to explore the details if desired.

Fourth, platforms should establish clear fairness guidelines and incorporate them into model development and evaluation. This may involve setting acceptable thresholds for performance disparities across geographic areas, and implementing fairness-aware optimization or post-processing techniques when these thresholds are exceeded. While such measures may reduce overall accuracy, they align with societal expectations and may reduce the risk of regulatory intervention.

6. Conclusion

6.1 Summary of Contributions

This research has provided a comprehensive investigation into the impact of artificial intelligence on predicting surge pricing in ride-hailing applications. Through the development and evaluation of seven predictive models across a large, multi-city dataset, the study has demonstrated that deep learning architectures, particularly transformer-based models, significantly outperform traditional forecasting methods in predicting dynamic price adjustments. The transformer achieved a 33.5% reduction in root mean squared error compared to baseline time-series models, representing a meaningful improvement with positive implications for passenger wait times, driver utilization, and platform efficiency.

However, these advances are accompanied by important limitations and trade-offs that must be carefully managed. The documented degradation in performance during novel demand shocks highlights the vulnerability of AI systems to distributional shifts, suggesting that autonomous AI deployment without human oversight is imprudent, particularly in emergency contexts. The fairness analysis revealed systematic disparities in prediction accuracy and surge exposure across income and racial lines, with the most accurate models exhibiting the largest disparities. This fairness–accuracy trade-off presents a challenging design dilemma for platform operators and a legitimate concern for regulators. Finally, the user perception findings indicate that the

most accurate models are not necessarily the most trusted, as interpretability and transparency are valued by passengers, sometimes at the expense of optimal predictive performance.

The study makes several contributions to the academic literature. It provides a comprehensive, methodologically rigorous comparison of AI-based surge prediction methods, establishing benchmark performance levels and identifying the relative strengths and weaknesses of each approach. It extends the empirical evidence base on algorithmic fairness in transportation, documenting disparities that have previously been assumed but not systematically measured. It contributes to the growing literature on AI interpretability by offering user-centered evidence on the value of transparent explanations. Finally, it bridges the gap between technical forecasting research and policy discourse, providing evidence-informed recommendations for regulatory frameworks.

6.2 Practical Implications

For ride-hailing platforms, the findings suggest that investment in AI-based surge prediction is warranted, but with careful attention to model selection, fairness monitoring, and interpretability infrastructure. The XGBoost model emerged as a strong practical choice, offering most of the accuracy benefits of deep learning with substantially lower computational costs and better interpretability. Platforms should prioritize the development of fairness monitoring dashboards that track performance across socioeconomic groups and geographic areas, enabling early detection of disparities and rapid corrective action.

For passengers, the results imply that AI-optimized surge pricing may deliver shorter wait times and more consistent availability, particularly during predictable peak periods. However, passengers should also be aware that surge predictions are less reliable during novel or extreme events, and that AI systems may reflect or amplify existing geographic inequities. The demand for transparent explanations, expressed strongly by participants in this study, suggests that platforms should invest in user-friendly interfaces that communicate surge drivers clearly.

For regulators, the study supports a balanced approach that neither mandates fixed pricing nor grants platforms complete discretion. Requirements for transparency, fairness reporting, and emergency contingency planning appear justified, while specific caps or rate structures may unnecessarily constrain the efficiency benefits of dynamic pricing. The fairness disparities documented in this study argue for ongoing monitoring and, where persistent disparities are identified, regulatory intervention to require corrective measures.

6.3 Limitations of the Study

Several limitations of this research should be acknowledged. The dataset, while extensive, is limited to three U.S. cities and may not generalize to other geographic contexts, particularly in developing countries where ride-hailing markets exhibit different characteristics. The proprietary nature of some data sources precludes full replication, though the core methodology is described in sufficient detail to guide similar investigations.

The fairness analysis is limited to area-level characteristics and cannot assess individual-level disparities, as passenger-level demographic data are not available. The disparities observed

may thus reflect area-based factors rather than individual discrimination. Similarly, the user perception study employed simulated explanations rather than real system interactions; responses may differ when passengers are facing actual surge pricing decisions with financial consequences.

The novel demand shock analysis, while instructive, is based on a limited number of events (two) and may not capture the full range of exceptional circumstances that platforms might encounter. Future research with longer observation periods and more extreme events would strengthen the evidence base on model robustness.

6.4 Future Research Directions

The findings of this study open several avenues for future research. First, the fairness–accuracy trade-off merits deeper investigation, including the development and evaluation of fairness-aware optimization methods specifically tailored to ride-hailing contexts. Experiments with different fairness constraints and their effects on both predictive accuracy and downstream outcomes (wait times, driver earnings) would provide practical guidance for platform operators.

Second, the interpretability results suggest a need for research on optimal explanation formats and interfaces. Experiments comparing different explanation modalities—textual, visual, quantitative—would help identify the most effective strategies for building user trust and understanding. The role of user demographics in explanation preferences also warrants investigation.

Third, the robustness of AI models to novel events could be improved through approaches such as data augmentation, simulation-based training, and domain adaptation techniques. Research that develops and validates these methods in the surge pricing context would enhance the practical reliability of AI systems.

Fourth, longitudinal studies examining the evolution of surge pricing patterns and model performance over time would provide insight into the stability of AI predictions and the need for ongoing model updating. The ride-hailing market is dynamic, with changing driver populations, evolving passenger behavior, and competitive responses from other platforms; models that perform well today may degrade as market conditions shift.

Fifth, comparative studies across different ride-hailing platforms and regulatory regimes would illuminate how market structure and policy context influence surge pricing outcomes and fairness implications. Such comparative work would provide valuable evidence for international policy harmonization efforts.

6.5 Closing Remarks

The integration of artificial intelligence into ride-hailing surge pricing represents both a significant technical advance and a sociotechnical challenge that requires careful, balanced management. The predictive capabilities of AI are genuine and substantial, offering the promise of more efficient, responsive transportation systems that better serve both passengers

and drivers. Yet these capabilities are not without risks: AI systems can amplify inequities, fail in unpredictable ways, and undermine user trust through opacity.

The path forward lies not in whether to adopt AI for surge pricing, but in how to deploy it responsibly. Platforms, regulators, and researchers must work together to develop systems that are accurate, fair, transparent, and robust—a multi-dimensional optimization problem that is as much ethical and political as it is technical. This study contributes to that endeavor by providing empirical evidence on the performance, fairness, and interpretability of AI-based surge prediction, and by offering practical guidance for navigating the trade-offs inherent in these systems.

As ride-hailing continues to reshape urban mobility, the decisions made about algorithmic pricing will have lasting consequences for economic opportunity, social equity, and public trust in technology. The findings of this research suggest that with thoughtful design and responsible governance, AI can contribute positively to these outcomes, but that optimism must be tempered with caution, monitoring, and a commitment to the fairness and transparency that users rightfully demand.

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